MILITARY SPECIFICATION

PACKING, PREFORMED, PTFE (POLYTETRAFLUORCETHYLENE) GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers PTFE O-rings.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipboard and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-ESC-S, Picatinny Arsenal, NJ 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5330

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(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

ASTM

ASTM D 570 - Plastic, Water Absorption of

ASTM D 638 - Plastic, Tensile Properties of, Standard Test Methods for

ASTM D 695 - Plastics, Rigid, Compressive Properties of

ASTM D 792 - Plastics, Specific Gravity and Density of, by Displacement

(Applications for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Materials. O-rings manufactured in accordance with this specification shall be fabricated from 100 percent virgin PTFE (see 4.3.7).

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3.2.1 Color. Color shall be uniformly white with no noticeable streaky or blotchy appearance. A slight brownish tint shall not be considered detrimental. The surface shall have no dead chalklike appearance which is an indication of porosity or other detrimental condition. There shall be no noticeable grainy appearance.

3.3 Dimensions and tolerances. The dimensions and tolerances of O-rings procured to this specification shall conform to the dimensions of MIL-P-70341/1.

3.4 Physical properties. The physical properties of the PTFE shall conform to the requirements listed in Table I.

TABLE I. PHYSICAL PROPERTIES.

ORIGINAL PROPERTIES	REQUIREMENTS	TEST METHOD
Specific gravity	2.13 - 2.18	4.3.1
Tensile strength, psi, min.	1500	4.3.2
Elongation, %, min.	95	4.3.3
Compression strength, psi	450	4.3.4
Water absorption, %	0.000	4.3.5
Porosity	0.000	4.3.6

3.5 Identification All O-rings shall be individually packaged. Each envelope shall have the following information printed on the outside.

a. National Stock Number
b. Military Part Number
c. Material Specification
d. Manufacturer's Name/CAGEC
e. Manufacturer's Compound Number
f. Manufacturer's Batch Number
g. Contract Number

3.6 Workmanship. The finished O-rings shall be uniform in quality and free from burrs, marks, foreign matter or other defects that will affect the life, serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with the test conditions specified in 4.3 through 4.3.7 of this specification.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the test methods listed below.

4.3.1 Specific gravity. The specific gravity shall be in accordance with Table I and tested according to ASTM D 792.

4.3.2 Tensile strength. The tensile strength shall be in accordance with Table I and tested according to ASTM D 638.

4.3.3 Elongation. The elongation shall be in accordance with Table I and tested according to ASTM D 638.

4.3.4 Compression strength. The compression strength shall be in accordance with Table I and tested according to ASTM D 695.

4.3.5 Water absorption. The water absorption shall be in accordance with Table I and tested according to ASTM D 570.

4.3.6 Porosity. The porosity shall be in accordance with Table I and tested according to the following method. The sample part shall be dip-or brush-coated with suitable red dye penetrant. The coat shall be allowed to stand for ten minutes and then be wiped off with a cloth saturated with solvent. Any retained dye other than minor indications attributable to surface orange peel molded finish or machining irregularities, is an indication of material porosity and shall be cause for rejection.

4.3.7 Material certification. The contractor shall provide to the government when requested certification verifying the use of proper materials defined in 3.2 and 3.2.1.

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4.4 Classification of defects. Defects shall be classified as identified in Table II.

TABLE II. CLASSIFICATION OF DEFECTS.

CATEGORY	DEFECT
Critical	
1	Dimensions not as specified
2	Cracks, burrs, foreign matter as defined in 3.6
Major	
101	Tensile strength
102	Elongation
103	Compression strength
104	Water absorption
105	Porosity
106	Color
Minor	None defined

4.5 Sampling for inspection. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, Inspection Level II, Acceptable Quality Level (AQL), 1.5 percent defective.

4.5.1 Inspection lot. A lot shall consist of all PTFE O-rings of the same size (dash number) made from the same batch of material and submitted for inspection at the same time and place.

4.5.2 Batch. A batch shall be defined as a quantity of material received from the resin manufacturer which has been assigned the same batch number.

4.6 Examination of defects in packaging. An examination shall be made to determine that the packaging, packing and markings comply with Section V. The sample unit shall be one shipping container fully packed selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be Level A (see 6.2).

5.2 Packing. Packing shall be Level B as specified (see 6.2.)

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5.3 Marking. Marking shall be in accordance with MIL-STD-129 unless otherwise specified in the procurement request.

5.4 Packaging inspection. The inspection of the packaging requirements shall be in accordance with MIL-STD-105.

6. NOTES

6.1 Intended use. O-rings procured in accordance with this specification are intended for use as static seals at temperatures ranging from -320°F to +450°F.

6.2 Ordering data.

6.2.1 Acquisition requirements. Procurement documents shall specify the following:

a. Title, number and date of this specification
b. Sizes in accordance with MIL-P-70341/1
c. Quantity
d. Federal stock number
e. Applicable levels of preservation, packaging, packing and marking required (see 5.1 through 5.4)

6.3 Key words.

a. Polytetrafluoroethylene
b. Packing
c. Preformed
d. PTFE

Custodians:
Army - AR
Navy - SH
Air Force - 99

Preparing Activity:
Army - AR

Agent:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-P-70341	

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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